



OPEN

Low cost and compact six switch seven level grid tied transformerless PV inverter

N. P. Gopinath¹, R. Azhagumurugan², M. Jagabar Sathik³✉ & D. Kirubakaran¹

Transformerless inverters with common ground structure are favoured in grid-connected photovoltaic (PV) systems primarily due to their ability to effectively suppress leakage current, eliminate transformer-related losses, enhance efficiency, and reduce costs. A six switch seven-level (S2-7 L) common ground type triple boost transformerless inverter topology for grid-tied solar PV applications is presented in this paper. The proposed structure maintains a constant common-mode voltage by sharing a common ground point between the source and the grid neutral, so effectively suppressing leakage current. The proposed topology's key advantages include generating a seven-level output voltage with only six switches, minimal conducting switches, and the lowest total standing voltage per unit. The article includes the modes of operation, design of passive components, and filter inductors. The MATLAB/Simulink software is employed for the simulation and experimental validation of 825 W through the use of a prototype model. A comprehensive comparison analysis of the proposed topology with the existing 7-level SC topologies is also included. The FFT analysis shows that the S2-7 L inverter has a THD of 17.2% for voltage, and 3.9% for grid current, confirming that these values are within the limits specified by the IEEE-519 standards.

Keywords Common ground, Inverter, Switched capacitor, Transformerless, Multilevel

Multilevel inverters (MLIs) are critical in integrating renewable energy sources with the grid since the varying forms of DC power from sources like the sun and wind will have to be converted to fixed forms of AC power which are compatible with the grid. Research in recent years has aimed to mitigate the limitations of conventional two-level and three-level MLIs, such as high switching losses, excessive THD, switch voltage stress, limited output waveform quality, a high number of components, capacitor balancing issues, and the need for costly, bulky filters. These limitations have driven the development of new multilevel topologies aimed at reducing these drawbacks and enhancing MLI adaptability for applications in renewable energy integration with grid infrastructure¹. Now-a-days SC-based inverters have become increasingly popular as a solution to the limitations of traditional two-level and three-level inverters. These use the series and parallel connection concept with the source and achieve higher output voltage levels with enhanced voltage boosting compared to traditional MLIs. Enhanced voltage boosting, reduced THD, lower voltage stress on switches, compact design, high efficiency, flexible voltage levels based on number of SCs, and self-voltage balancing make switched-capacitor inverters ideal for modern power conversion, particularly in renewable energy integration². In the literature, several SC-based multilevel inverters (MLIs) have been proposed to generate higher output voltage levels, such as 5-level, 7-level, 9-level, and 13-level, with both boosting and non-boosting capabilities^{3–29}.

The authors in^{3,4} propose 5-level inverter topologies utilizing switched capacitors (SCs). In³, a 5-level output voltage is obtained using two SCs; however, it does not achieve voltage boosting. Conversely, the inverter topology presented in⁴ achieves dual voltage boosting using two SCs and eight power switches. Nevertheless, this improvement requires that 4 of the 8 switches be rated to withstand the output voltage level. The inverter topologies presented in^{5–25} employ SCs in various 7-level designs, each with its own set of pros and cons. The topologies presented in^{7,13,14,16,19,23} are 7-level inverter designs with a voltage gain of 1.5 times the input voltage. In contrast, the topologies in^{5,6,8–12,15,17,18,20–22,24} are capable of producing a threefold voltage boost. In⁵, the authors presented a 7-level inverter topology using three SCs. The SCs are rated at V_{dc} , $2V_{dc}$, and $3V_{dc}$ to achieve triple voltage boosting. The same authors presented another 7-level inverter with the common ground feature in⁶, which uses one additional power switch and one SC lesser than the⁵. Further, the peak inverse voltage

¹Department of Electrical and Electronics Engineering, St. Joseph's Institute of Technology, Chennai, India.

²Department of Electrical and Electronics Engineering, Sri Sai Ram Engineering College, Chennai, India.

³Department of Electrical and Electronics Engineering, Faculty of Engineering and Technology, SRM Institute of Science and Technology, Kattankulathur, Chengalpattu 603203, Tamil Nadu, India. ✉email: mjsathik@ieee.org

of⁶ is less than the output voltage, whereas it is equal to the output voltage in⁵ for 3 out of its nine switches⁷. proposes another seven-level inverter with two switched capacitors (SCs) and nine switches. However, it has a relatively high total standing voltage per unit (TSV_{p.u.}) along with limited voltage boosting capability. Although these topologies achieve threefold voltage boosting, their TSV_{p.u.} remains significantly high. Additionally, topology⁹ incorporates an H-bridge in its structure. In^{10,11}, 7-level topologies with nine power switches are presented. Similar to topologies^{8,9}, these designs also exhibit a high TSV_{p.u.}. Furthermore, topology¹¹, similar to⁹, incorporates an H-bridge. Due to the inclusion of an H-bridge, during the switching operation, each of the switches in the H-bridge must withstand the full supply voltage when in the off-state. This leads to a high blocking voltage across the switches. Although the topology presented in¹² achieves a threefold voltage gain, it requires a higher number of power switches to do so. This added switch count can increase the circuit's complexity, cost, and overall power loss, potentially affecting efficiency. The topologies in^{13,14} use ten power switches with one switched capacitor (SC) and eight power switches with two SCs, respectively. In addition to their limited voltage gain of 1.5, their ANPC structure is less effective in suppressing leakage current compared to a common-ground configuration^{5,6,9,10}. Few other 7-level inverter topologies in the literature have drawbacks, i.e., topology¹⁵ requires a high number of switches due to its H-bridge configuration, while topology¹⁶ shows limited voltage gain despite a high component count and has an excessively high TSV_{p.u.}. Additionally, topologies^{17–19} use a higher number of switches and are also restricted to a modest voltage gain. Similar topologies with the same number of switches but differing in the number of switched capacitors (SCs) to achieve a threefold voltage boost are presented in^{20,21}. However, these designs suffer from a high TSV_{p.u.} while using more power switches, significant inrush currents and leakage current issues limit their practical application. In^{22,23}, reduced-switch 7-level topologies are presented, achieving threefold and 1.5x voltage boosts, respectively. Topology²³ uses only eight switches to generate a 7-level output; however, it is limited by a low voltage gain and a TSV_{p.u.}. In^{25–27}, SC-based inverter topologies for 9-level voltage generation, and in^{28,29}, for 13-level generation, are presented. While these topologies achieve voltage boosting, they adhere to the traditional trade-off: configurations with more switches have a low TSV_{p.u.}, whereas those with fewer switches exhibit a high TSV_{p.u.}.

Driven by the challenge of designing topologies with fewer power switches with low TSV while achieving high output voltage levels, including voltage boosting and common-ground capability to suppress leakage current, this paper introduces a new 7 L inverter topology. Key features of the proposed S2-7 L inverter topology include:

- (1) Reduced power switches, i.e., only six switches
- (2) Low TSV_{p.u.} Of value 4.67
- (3) The common ground feature that suppresses leakage current
- (4) Sensorless switched capacitor voltage balancing
- (5) Cost-effective with fewer power components and no capacitor voltage balancing or sensors

Proposed seven-level triple boost inverter topology

Circuit demonstration

The proposed inverter topology is configured using a single source, 7 switches (P_1 - P_7), four SCs (C_a - C_d), and four diodes (D_a - D_e), as shown in Fig. 1. It has two units: the source unit (SU) and the SC unit (SCU). The SU is formed using a single source, two switches (P_1 - P_3), diode D_a , and SC C_a . The SCU comprises two switches (P_3 and P_4), and two diodes (D_c and D_d). The voltage across the SCs V_{Ca} and V_{Cb} is self-balanced at V_{dc} , and the SCs V_{Cc} and V_{Cd} are balanced at $2V_{dc}$, respectively. The proposed inverter topology can generate a seven-level output voltage with boost factor 3 at the inverter output terminals by utilizing the source voltage and SC voltages (V_{Ca} - V_{Cd}). The proposed inverter topology can suppress the leakage current ~ 0 due to their direct ground sharing point connection between the source negative and grid neutral.

Description of operation levels

The seven-level output voltage waveform of the proposed S2-7 L is generated using the switch sequences given in Table 1. Figure 2(a) shows the maximum blocking voltage (MBV_{p.u.}) of switches V_{P1} - V_{P6} of the proposed S2-7 L inverter topology. It can be seen that the MBV of the switches P_1 and P_2 is equal to V_{dc} , P_3 and P_4 is equal to $2V_{dc}$, and the remaining two switches P_5 and P_6 are equal $4V_{dc}$. Using MBV_{p.u.} of switches, the total standing voltage per unit of the proposed inverter topology is calculated in (1) as follows,

$$TSV_{p.u.} = \sum_{z=1}^6 V_{S_z} / 3V_{dc} = \frac{14V_{dc}}{3V_{dc}} = 4.67 \quad (1)$$

Further, Fig. 2(b) to (h) depicts the active switches, capacitor charging and discharging, and load current path while generating seven-level output voltage waveforms. Each of the seven distinct output voltage levels is described as follows:

Level I

The output voltage $v_o = +V_{dc}$ is obtained via the PV source due to the active switch P_5 . Further, the switches P_2 and P_4 are also switched ON to charge the SCs C_a and C_c to V_{dc} and $2V_{dc}$, via $a-D_a-P_2-n$ and $a-D_a-D_c-P_4$ as depicted in Fig. 2(b).

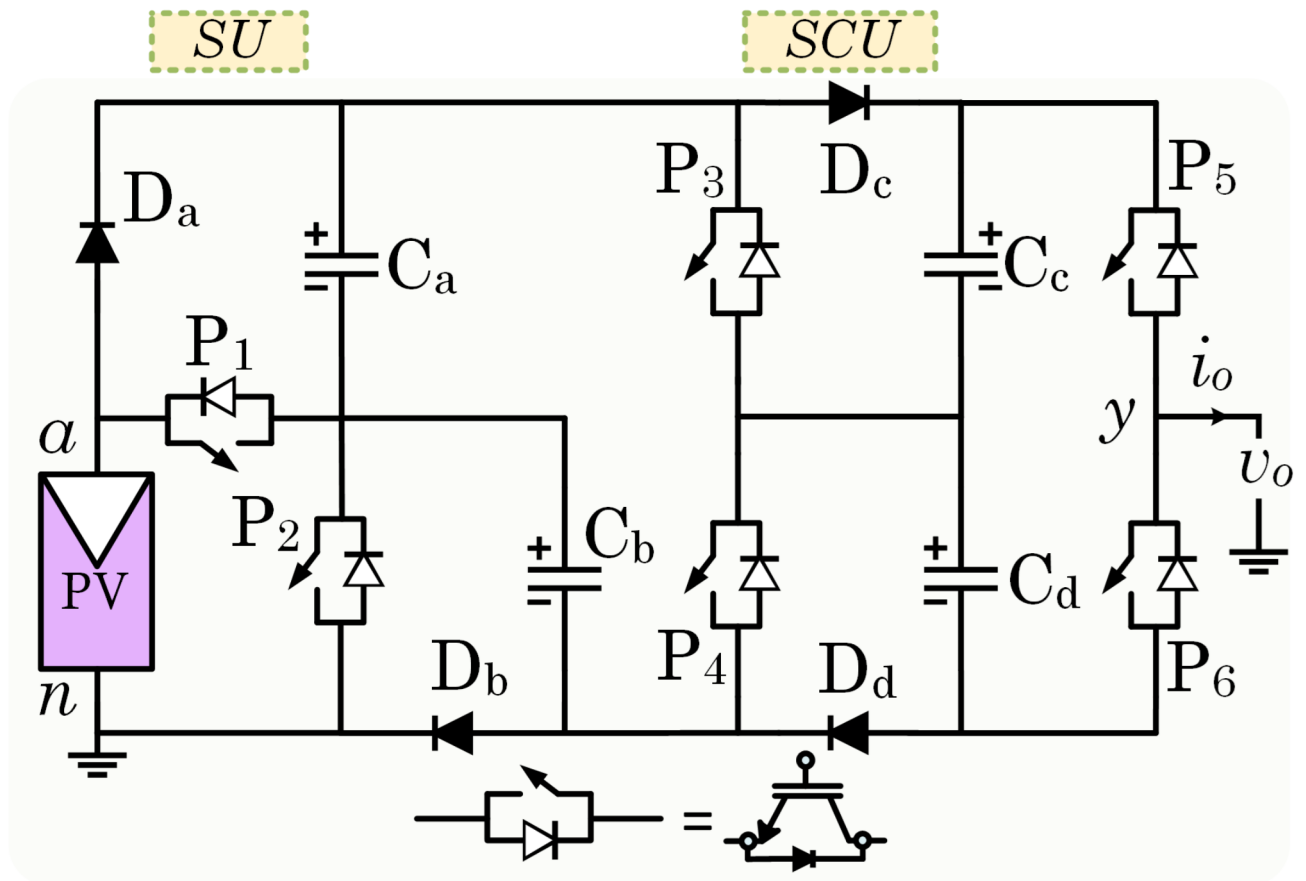


Fig. 1. Circuit diagram of Proposed S2-7 L topology.

Output Voltage (v_o)	P_1	P_2	P_3	P_4	P_5	P_6	C_a	C_b	C_c	C_d
$+V_{dc}$	0	1	0	1	1	0	↑	↓	↑	-
$+2V_{dc}$	1	0	0	0	1	0	↓	↑	↑	-
$+3V_{dc}$	0	1	1	0	1	0	↑	-	↓	-
0	1	0	1	0	0	1	-	↑	-	↑
$-V_{dc}$	0	1	1	0	0	1	↑	↓	-	↑
$-2V_{dc}$	1	0	0	1	0	1	-	↑	-	↓
$-3V_{dc}$	0	1	0	1	0	1	↑	↓	-	↓

Table 1. Switching sequence of S2-7 L. 1&0=On&Off Level of Switch, ↑=Charging, ↓=Discharging, -=No Effect

Level II

The PV source and the SC C_a are connected in series to generate the output voltage of $v_o = +2V_{dc}$. As shown in Fig. 2(c), the switches P_1 and P_5 connect the terminals 'a' and 'y'. The SC C_b is charged to a voltage of V_{dc} because the input source is connected in parallel via a - P_1 - D_b - n .

Level III

The terminal denoted as 'y' and the positive terminal 'a' of the PV source is connected through the C_c and the switches P_3 and P_5 to obtain the output voltage of $v_o = +3V_{dc}$. The switched capacitor C_a is charged to a voltage of V_{dc} because the input source is connected in parallel via a - D_a - P_2 - n as shown in Fig. 2(d).

Level IV

The switch P_6 is turned ON during this Level, shorting the terminals 'n' and 'y' as shown in Fig. 2 (e). Since the output terminals are shorted, the output voltage in this Level is $v_o = 0$. The SC C_b is charged to V_{dc} via the a - P_1 - D_b - n , and the other two SCs as shown in Fig. 2(e).

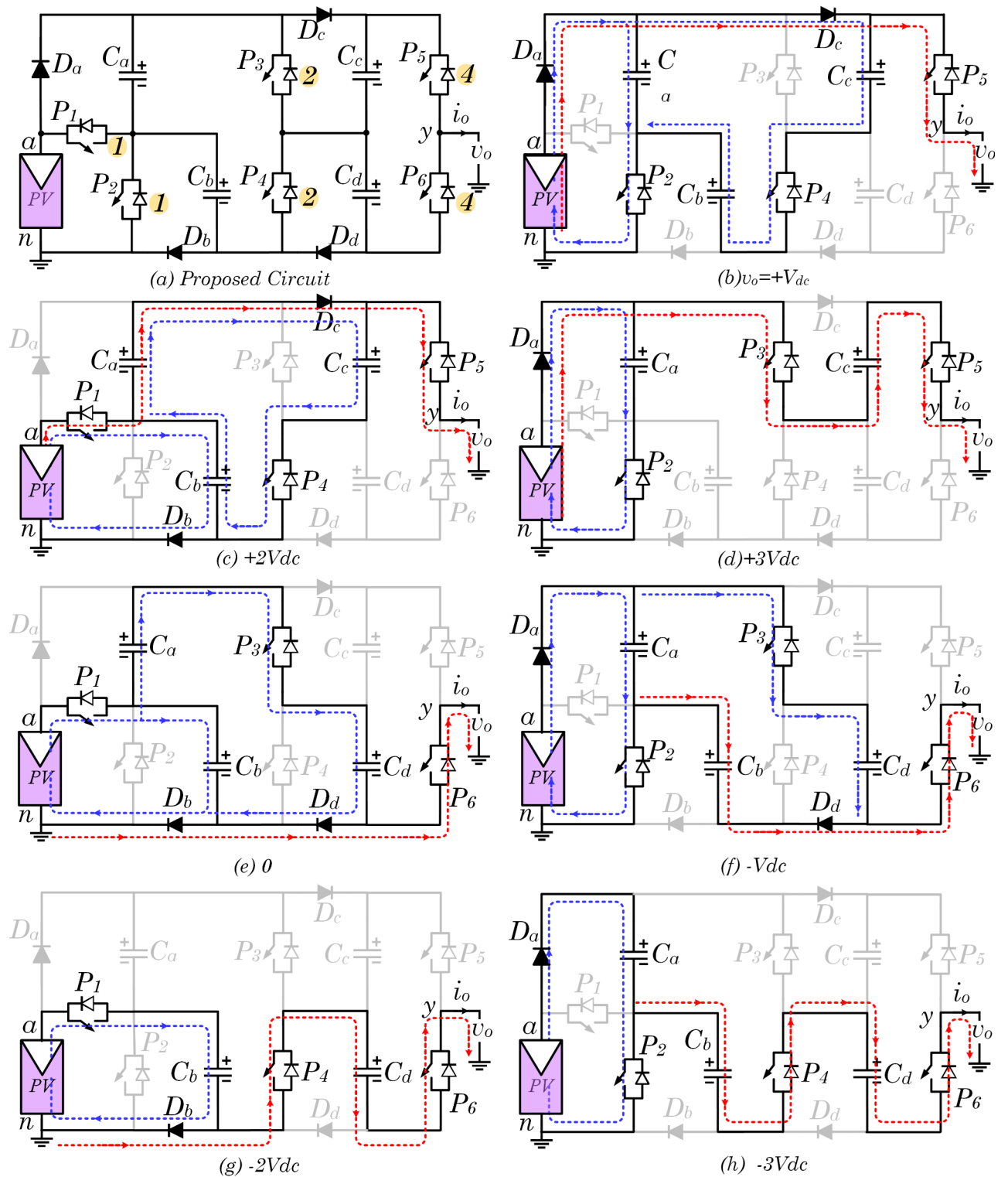


Fig. 2. (a) $MBV_{p.u.}$ of the proposed S2-7 L topology, (b)-(h) Output voltage levels from $+3V_{dc}$ to $-3V_{dc}$.

Level V

The first negative voltage level generation Level i.e., $v_o = -V_{dc}$, is achieved due to the discharging of the SC C_b . The switches P_2 and P_3 are switched ON to charge the SCs C_a and C_d to V_{dc} and $2V_{dc}$, respectively, via a - D_a - P_2 - n and a - D_a - P_3 - D_d as illustrated in Fig. 2(f).

Level VI

In this Level, the switches P_4 and P_6 are switched ON; thus the negative terminal of the SC C_d is connected with 'y'. Hence, the SC C_d discharges to obtain the load voltage of $v_o = V_{Cd} = -2V_{dc}$. The SC C_b is charged to V_{dc} via a - P_1 - D_b - n , as shown in Fig. 2(g).

Level VII

Here the SCs C_b and C_d discharge to generate the output voltage of $v_o = V_{Cb} + V_{Cd} = -3V_{dc}$.

In addition to the active switches P_4 and P_6 , to obtain $-3V_{dc}$, the switch P_2 is also turned ON to charge the SC C_a to V_{dc} , as shown in Fig. 2(h).

Design of SCs (C_a - C_d) and filter inductor (L_f)

The PWM method and seven-level output voltage waveform of the proposed S2-7 L topology are shown in Fig. 3. The charging and discharging status of capacitors C_a - C_d during all 7 L output voltage (v_o) levels are listed in Table 1. The voltage across capacitors has to be balanced to ensure constant output voltage at terminals. The longest discharge time of capacitors (LDT), a permissible voltage ripple of 10%, and the operating frequency of the inverter structure are all factors that can be considered to achieve this objective. Considering the LDT for C_a is t_x to $\pi - t_x$, C_b & C_c is t_y to $\pi - t_y$, and C_d is $\pi + t_y$ to $2\pi - t_y$ from Fig. 2(b) to (h), the capacitance of C_a - C_d is calculated as follows (2)-(5),

$$C_a = \frac{I_{gmx}}{\omega \times \% \text{Ripple} \times V_{dc}} [\cos(\omega t_x) - \cos(\omega(\pi - t_x))] \quad (2)$$

$$C_b = \frac{I_{gmx}}{\omega \times \% \text{Ripple} \times V_{dc}} [\cos(\omega t_y) - \cos(\omega(\pi - t_y))] \quad (3)$$

$$C_c = \frac{I_{gmx}}{\omega \times \% \text{Ripple} \times V_{dc}} [\cos(\omega t_y) - \cos(\omega(\pi - t_y))] \quad (4)$$

$$C_d = \frac{I_{gmx}}{\omega \times \% \text{Ripple} \times V_{dc}} [\cos(\omega t_x) - \cos(\omega(\pi - t_x))] \quad (5)$$

The Eq. (6) represents the current through the filter inductor over a full cycle,

$$i_{L_f}(t) = i_{L_f}(0) + \frac{1}{L_f} \int_0^t V_{L_f} dt \quad (6)$$

From Eq. (6), the value of L_f can be calculated as (7),

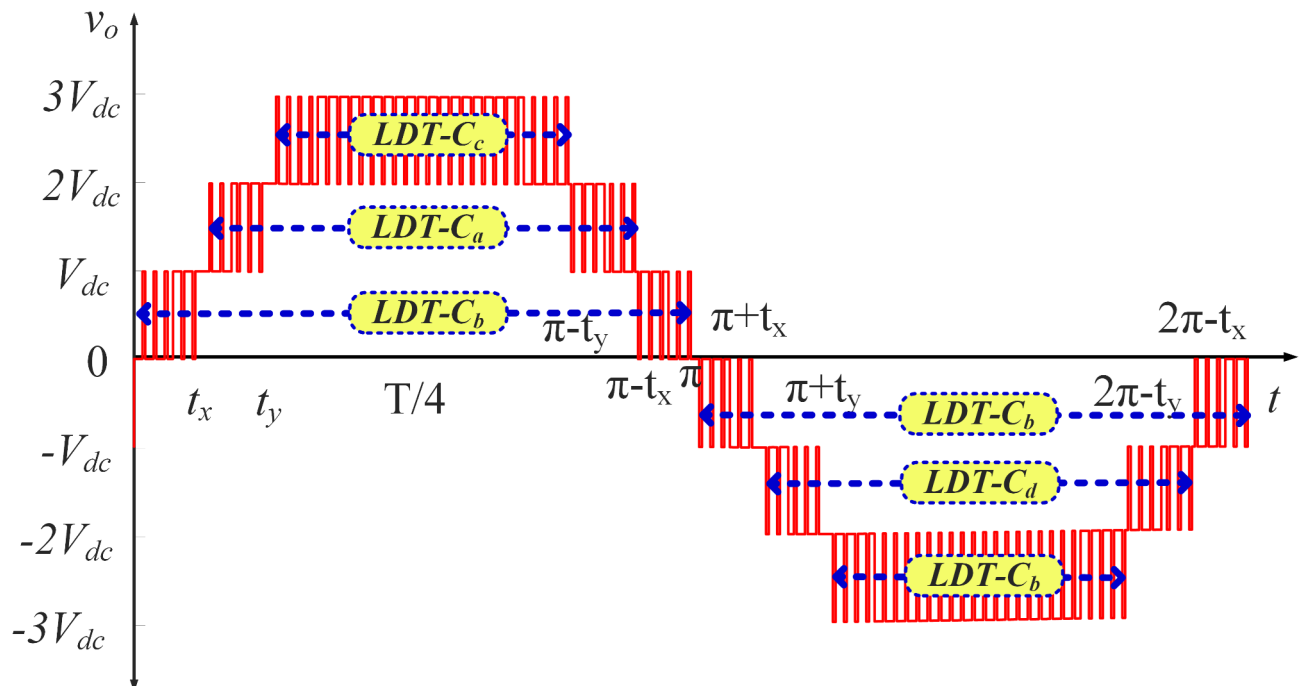


Fig. 3. LDT of SCs C_a - C_d and S2-7 L output voltage waveform.

$$L_f = \frac{1}{f_{sw} \times \Delta i_{L_f}} \left[5V_m \sin \omega t - 6V_{dc} - \frac{V_m^2 \sin^2 \omega t}{V_{dc}} \right] \quad (7)$$

The inductor filter (L_f) value with peak inductor current ripple is calculated as (8),

$$L_f = \frac{1}{f_{sw} \times \Delta i_{L_f, max}} \left[5V_m - 6V_{dc} - \frac{V_m^2}{V_{dc}} \right] \quad (8)$$

Power loss analysis

The total power loss of the proposed S2-7 L topology is calculated in this section.

Conduction loss

The conduction losses, is determined by factors such as the on-state resistances of the switch ($R_{n,p}$), on-state resistances of the diode ($R_{n,D}$), and the equivalent series resistance ($R_{esr,C}$).

$$\begin{aligned} P_{C+1} &= I_{P2}^2 R_{n,P2} + I_{P4}^2 R_{n,P4} + I_{P5}^2 R_{n,P5} + I_{D1}^2 R_{n,D1} + I_{D2}^2 R_{n,D2} + I_{C1}^2 R_{esr,C1} + I_{C2}^2 R_{esr,C2} + I_{C3}^2 R_{esr,C3} \\ P_{C+2} &= I_{P1}^2 R_{n,P1} + I_{P4}^2 R_{n,P4} + I_{P5}^2 R_{n,P5} + I_{D2}^2 R_{n,D2} + I_{D3}^2 R_{n,D3} + I_{C1}^2 R_{esr,C1} + I_{C2}^2 R_{esr,C2} + I_{C3}^2 R_{esr,C3} \\ P_{C+3} &= I_{P2}^2 R_{n,P2} + I_{P3}^2 R_{n,P3} + I_{P5}^2 R_{n,P5} + I_{D1}^2 R_{n,D1} + I_{D2}^2 R_{n,D2} + I_{C1}^2 R_{esr,C1} + I_{C2}^2 R_{esr,C2} \\ P_{C_{zero}} &= I_{P1}^2 R_{n,P1} + I_{P3}^2 R_{n,P3} + I_{P6}^2 R_{n,P6} + I_{D2}^2 R_{n,D2} + I_{D3}^2 R_{n,D3} + I_{C1}^2 R_{esr,C1} + I_{C2}^2 R_{esr,C2} \\ P_{C-1} &= I_{P2}^2 R_{n,P2} + I_{P3}^2 R_{n,P3} + I_{P6}^2 R_{n,P6} + I_{D1}^2 R_{n,D1} + I_{D2}^2 R_{n,D2} + I_{C1}^2 R_{esr,C1} + I_{C2}^2 R_{esr,C2} + I_{C3}^2 R_{esr,C3} \\ P_{C-2} &= I_{P1}^2 R_{n,P1} + I_{P4}^2 R_{n,P4} + I_{P6}^2 R_{n,P6} + I_{D2}^2 R_{n,D2} + I_{D3}^2 R_{n,D3} + I_{C1}^2 R_{esr,C1} + I_{C2}^2 R_{esr,C2} \\ P_{C-3} &= I_{P2}^2 R_{n,P2} + I_{P4}^2 R_{n,P4} + I_{P6}^2 R_{n,P6} + I_{D1}^2 R_{n,D1} + I_{D2}^2 R_{n,D2} + I_{C1}^2 R_{esr,C1} + I_{C2}^2 R_{esr,C2} + I_{C3}^2 R_{esr,C3} \end{aligned} \quad (9)$$

The conduction losses for the proposed S2-7 L topology are computed for the all-output voltage levels using their equivalent circuits²⁸, as depicted in Fig. 4. Further, the total conduction loss is expressed in (10),

$$P_{T,C} = P_{C+3} + P_{C+2} + P_{C+1} + P_{C_{zero}} + P_{C-1} + P_{C-2} + P_{C-3} \quad (10)$$

Switching loss

Switching losses are the losses caused by the switching of semiconducting devices during turn-on and turn-off transitions¹⁵. Considering energy dissipation during the turn-on and turn-off periods,

$$P_{s,on,k} = \int_0^{T_{on}} V(t) \times I(t) dt = \frac{V_{sw,k} \times I_k \times T_{on}}{6} \quad (11)$$

$$P_{s,off,k} = \int_0^{T_{off}} V(t) \times I(t) dt = \frac{V_{sw,k} \times I'_k \times T_{off}}{6} \quad (12)$$

where I_k and I'_k are the currents flowing through the k^{th} switch at the time of switching ON and before switching OFF, respectively. $V_{sw,k}$ is the voltage of the switch in the OFF state. Using (11) and (12), the total switching loss is estimated using (13):

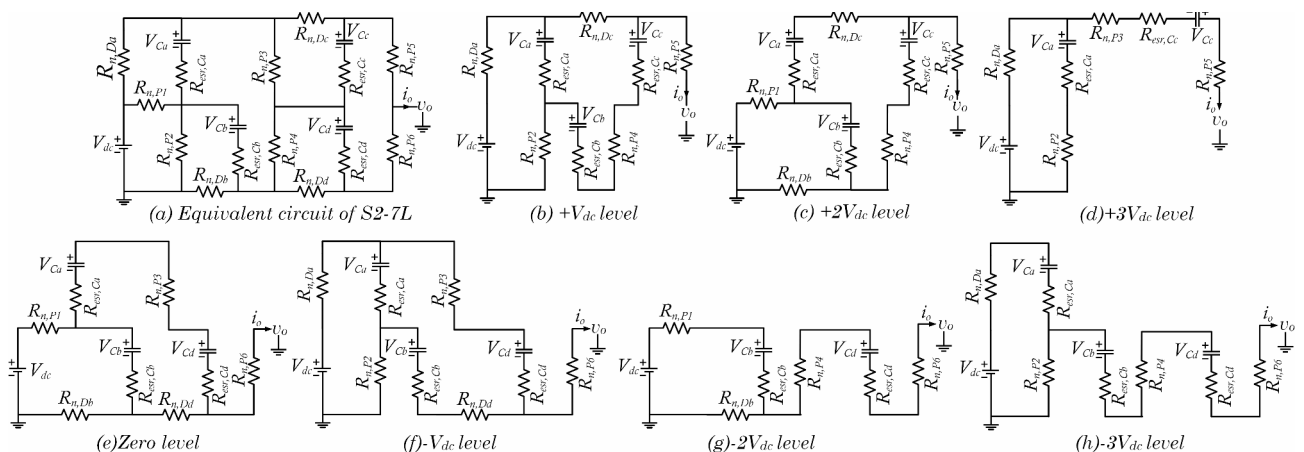


Fig. 4. Equivalent circuit of S2-7 L topology during different operating levels.

$$P_{SW} = \frac{1}{6T} \sum_{n=1}^6 \left(\sum_{j=1}^{N_{on}} P_{s,on,n,j} + \sum_{j=1}^{N_{off}} P_{s,off,n,j} \right) \tag{13}$$

where N_{on} and N_{off} denote the number of turns ON and turned OFF during one fundamental cycle.

Ripple loss

The ripple loss in the capacitor during its cyclic charging and discharging due to the variation of their voltages is expressed as,

$$P_{RIP} = f_o \left(\sum_{k=1}^4 C_k \times \Delta V_{Ck} \right) \tag{14}$$

The total loss can be computed through (8)–(14) as,

$$P_{T-Loss} = P_{T,C} + P_{SW} + P_{RIP} \tag{15}$$

The efficiency is calculated as (16),

$$\eta = P_o / P_o + P_{Loss} \tag{16}$$

Result and discussion

This section includes the performance of the proposed circuit topology by validating it using MATLAB/Simulink simulation tools and real-time implementation to conduct experimental testing on the proposed S2-7 L inverter topology. Furthermore, dynamic performance is tested under various load and step input variations, and the findings are explained.

Simulation results

The proposed S2-7 L was simulated in MATLAB/Simulink™ and PLEXIM simulation tools, and the respective results are presented in this section. The simulation and experimental parameters are listed in Table 2.

The proposed S2-7 L inverter topology is tested first with an R load of 100 Ω, producing in-phase output voltage and current waveforms, as displayed in Fig. 5(a). Subsequently, the inverter is tested with an RL load of 75 + j100 Ω, where the inductive element introduces a phase shift, causing the load current to lag the output voltage. The resulting waveforms, capturing both R and RL load conditions, are shown in Fig. 5(a) and 5(b), demonstrating the inverter’s behavior under different load types. It is seen that the proposed topology produces a threefold output voltage of 330 V when applying 110 V input voltage. The SC voltages associated with Fig. 5(c) are obtained when the input is set at 110 V, and as seen, the SC voltages V_{Ca} and V_{Cb} are balanced at 110 V while V_{Cc} and V_{Cd} at 220 V. These steady voltage values imply that balancing of voltage across the SCs was effective within the circuit and does not need any extra control methods.

Figure 6(a) depicts the inverter’s output voltage and load current when the input voltage is swept from 90 V to 110 V. Meanwhile, Fig. 6(b) displays the corresponding SC voltages (V_{Ca} – V_{Cd}) during this input variation. The results demonstrate the inverter’s response to input fluctuations, indicating how both the output voltage and current adapt to changing conditions. Furthermore, the load is adjusted between an R load of 100 Ω and an RL load of 75 + j100 Ω, with the respective results depicted in Fig. 6(c). These results showcase the dynamic response

Parameters		Simulation	Experiment
Input voltage (V_{dc})		110 V	
Grid voltage (v_g)		325 V	
Output power (P_o)		825 W	
Frequency	Switching frequency (f_{sw})	5 kHz	
	Output frequency (f_o)	50 Hz	
Switched capacitors	C_a	Alcon PG6DI 2200 μF-200 V	
	C_b	B43305A2278M2 2700 μF- 200 V	
	C_c	ALC70C272FP400 2700 μF – 400 V	
	C_d	ALC10C222FP400, 2200 μF – 400 V	
Filter inductor (L_f)		1.6 mH	
Soft charging (L_{sc})		33 μH	
Switch model		-	IKW50N60H3-600 V, 50 A IRGP4072DPBF-300 V, 40 A
Diode model		-	DPG60I300HA-400 V, 60 A
Controller		-	TMS320F28379D

Table 2. Parameters for S2-7 L inverter.

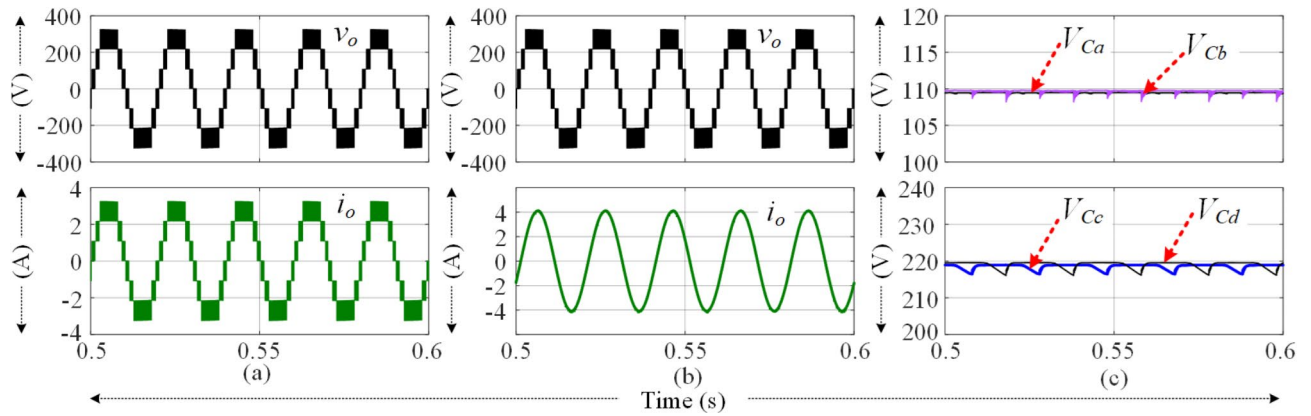


Fig. 5. Simulation results of proposed S2-7 L topology (a) v_o and i_o at $R=100\ \Omega$, (b) v_o and i_o at $Z=75+j100\ \Omega$, and (c) SC voltages V_{Ca} , V_{Cb} , V_{Cc} and V_{Cd} .

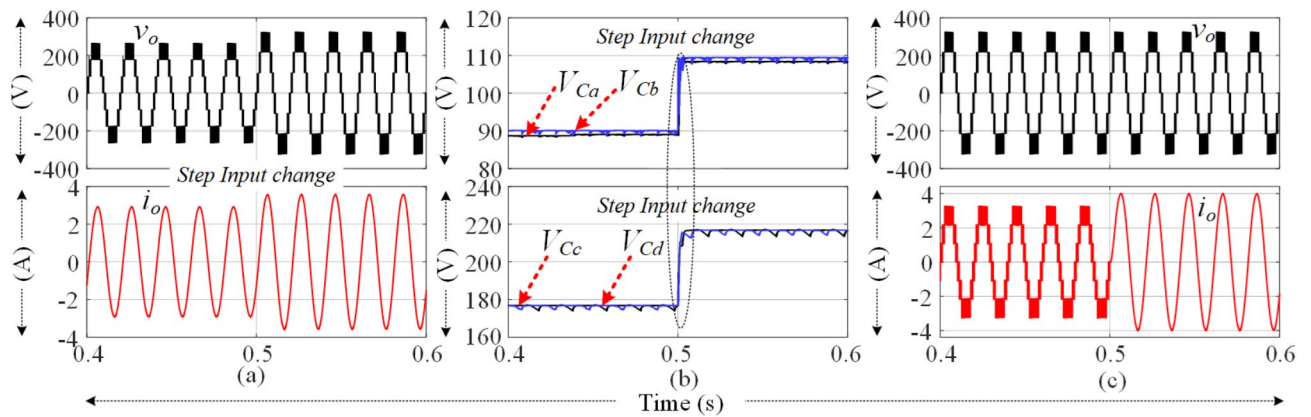


Fig. 6. Simulation results of proposed S2-7 L topology at (a) v_o and i_o at step variation of input from 90 V to 110 V, (b) Capacitor voltages during input change, and (c) v_o and i_o during load change from $100\ \Omega$ to $Z=75+j100\ \Omega$.

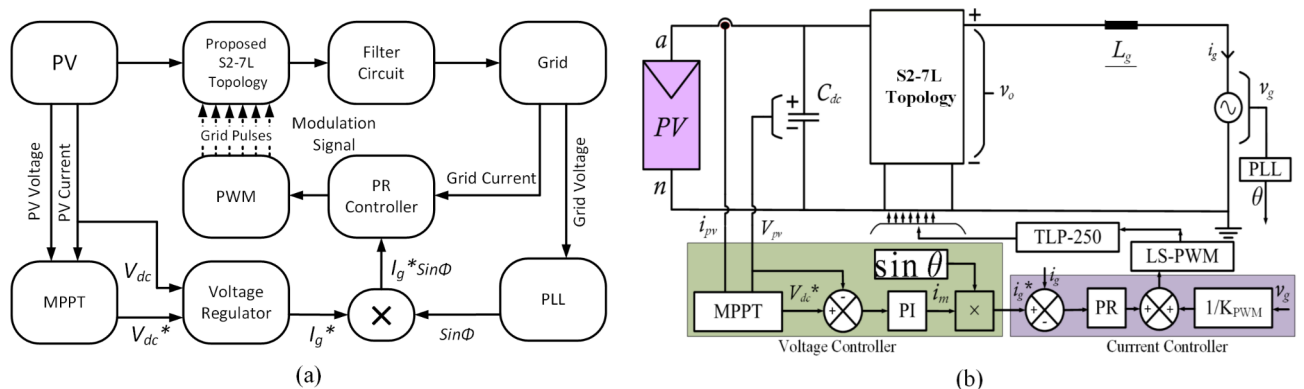


Fig. 7. (a) Block diagram of proposed S2-7 L topology, (b) Closed loop method of proposed S2-7 L topology.

behavior of the proposed S2-7 L topology under varying load conditions. The operation of the proposed topology in connection with the grid is assessed using a closed-loop control scheme that incorporates a PR controller⁶, as shown in Fig. 7.

Figure 8(a) shows the grid voltage (v_g), current (i_g), and inverter output (v_o) voltage waveforms when operating at a $\cos\phi=1$ and feeding real power of 800 W to the grid. In Fig. 8(a) and 8(b) the grid current (i_g) is multiplied by 30 to match the grid voltage. Moreover, the performance of the proposed topology at a power factor of 0.6 is

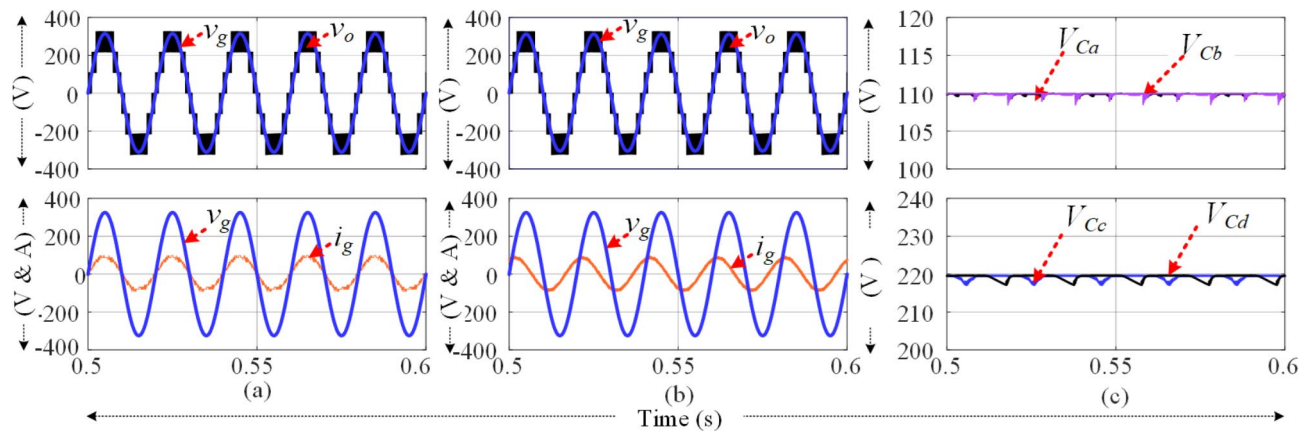


Fig. 8. Simulation results of proposed S2-7 L topology at (a) $\cos \phi = 1$, (b) $\cos \phi = 0.6$ Lead, and (c) SC voltages V_{Ca} , V_{Cb} , V_{Cc} and V_{Cd} at $\cos \phi = 1$.

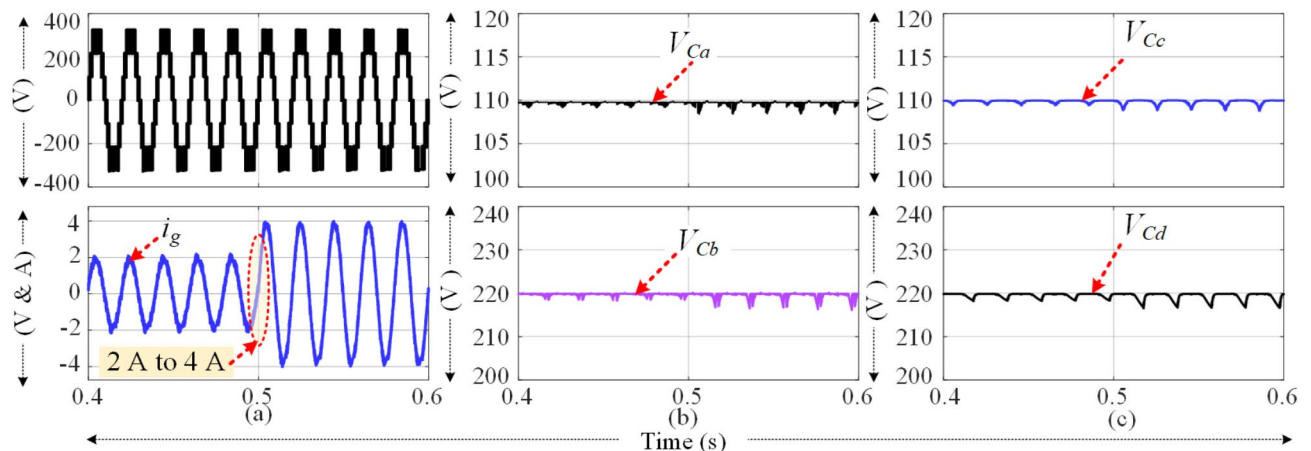


Fig. 9. Simulation waveforms of proposed S2-7 L topology when reference grid current ($i_{g,ref}$) is changed from 2 A to 4 A at $\cos \phi = 1$.

depicted in Fig. 8(b), while Fig. 8(c) illustrates the SC voltage waveforms of V_{Ca} - V_{Cd} in grid-connected mode. Moreover, as the reference grid current (i_{ref}) is varied from 2 A to 4 A at $\cos \phi = 1$, the operability of the proposed inverter is confirmed through the waveforms illustrated in Fig. 9(a). Simultaneously, the SC voltages V_{Ca} - V_{Cd} are presented in Fig. 9(b) and 9(c). This operation of the proposed topology is particularly relevant for solar PV applications, demonstrating its potential for seamlessly integrating renewable energy into the grid.

Experimental results

To investigate the feasibility of the proposed S2-7 L topology, a laboratory prototype of 825 W has been made as shown in Fig. 10, and several tests have been performed similar to simulation tests. The switch models IKW50N60H3 and IRGP4072DPbF, the diode model DPG60I300HA, and the controller TL28379D, along with the driver TLP driver circuit, were used to construct the proposed S2-7 L inverter topology. The experimental results are obtained for the input voltage of $V_{dc} = 110$ V, RMS value of the grid voltage $v_g = 230$ V, SCs (V_{Ca} - V_{Cd}) and filter inductor (L_f) values as per Eqs. (2)-(5) and (8) as C_a & $C_d = 2300$ μ F, C_b & $C_c = 2700$ μ F, and $L_f = 1.6$ mH. Figure 11(a) depicts the experimental responses of the inverter output voltage (v_o), load current (i_o), and SC voltage (V_{cd}) for a resistive load of $R = 100$ Ω . In contrast, Fig. 11(b) presents the results with a load of $RL = 75 + j100$ Ω .

The experimental waveforms reveal the inverter performs under different load conditions, emphasizing its ability to manage both resistive and reactive loads. The stable responses in Fig. 11(a) suggest effective operation with the purely resistive load, while Fig. 11(b) demonstrates the inverter's adaptability to the reactive component, which influences the voltage and current waveforms. Figure 12 (a) and (b) show the result of varying the input voltage (V_{dc}) from 90 V to 110 V at $RL = 75 + j100$ Ω . As the input voltage is varied, the voltages of the SC (V_{Cc} and V_{Cd}) remain balanced at ~ 220 V. While varying the input voltage the output voltage is 270 V to 330 V verifying its dynamic operability of the proposed topology.

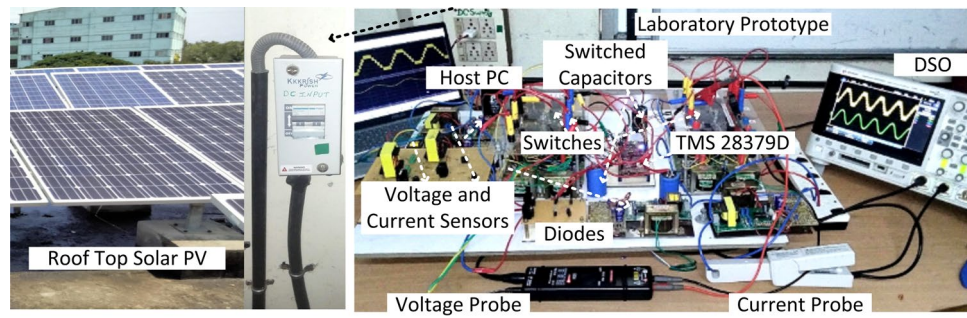


Fig. 10. Laboratory prototype of S2-7 L.

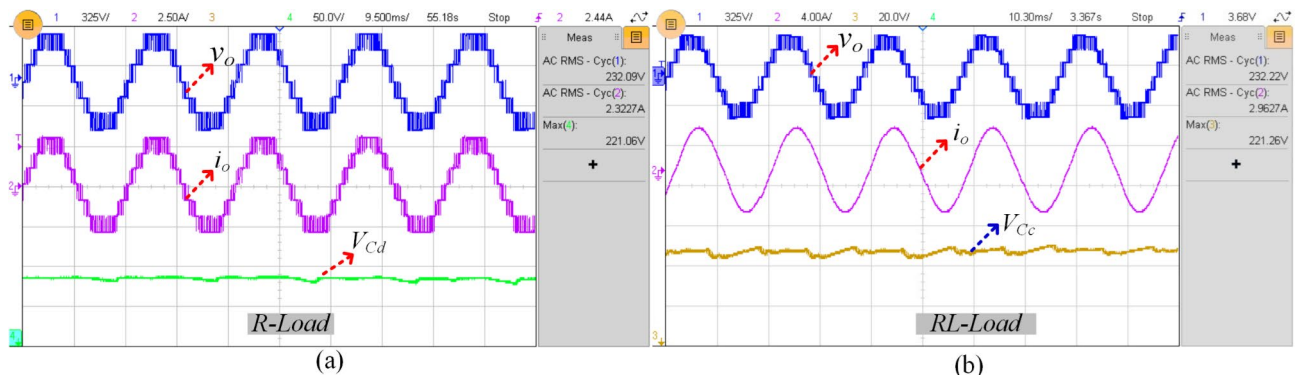


Fig. 11. Experimental results of proposed S2-7 L topology (a) v_o , i_o , and V_{cd} at $R=100\ \Omega$, (b) v_o , i_o , and V_{cc} at $Z=75+j100\ \Omega$.

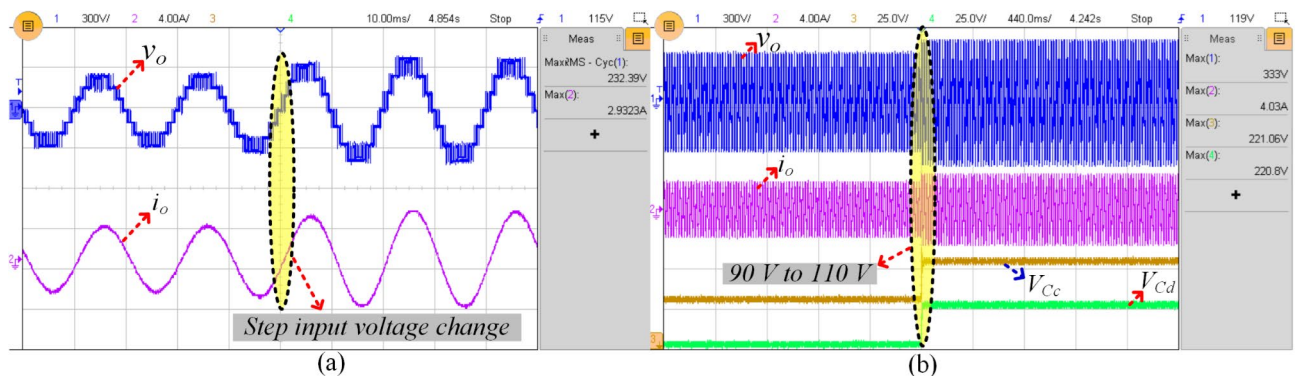


Fig. 12. Experimental results of proposed S2-7 L topology When input is varied from 90 V to 110 V (a) v_o and i_o at $Z=75+j100\ \Omega$, (b) v_o , i_o and V_{cd} at $Z=75+j100\ \Omega$.

Further load variations have been considered from RL load to R load and the respective results of output voltage (v_o), load current (i_o), and SC voltage (V_{cd}) are depicted in Fig. 13(a). Also, modulation index values is changed from 1 to 0.7 and then 0.7 to 0.4 is considered and the corresponding results are shown in Fig. 13 (b) and (c). All these results verify the dynamic operation capability of the proposed S2-7 L inverter topology under different transient operation conditions.

The grid voltage (v_g), current (i_g), and inverter output voltage (v_o) while injecting a grid current of $\sim 3.6\text{ A}$, as shown in Fig. 14(a). It is seen that seven levels of output voltage are three-fold, i.e., 330 V, achieved at unity power factor condition (UPF) while the input is 110 V. The operation of the proposed topology is further tested with 0.6 lagging power factor, and the respective inverter output voltage (v_o), along with grid voltage (v_g), and current (i_g), are illustrated in Fig. 14(b). Further to verify the performance during the dynamic operating condition, the reference current is boosted from 1.5 A to 3 A and the respective waveforms are depicted in Fig. 14(c).

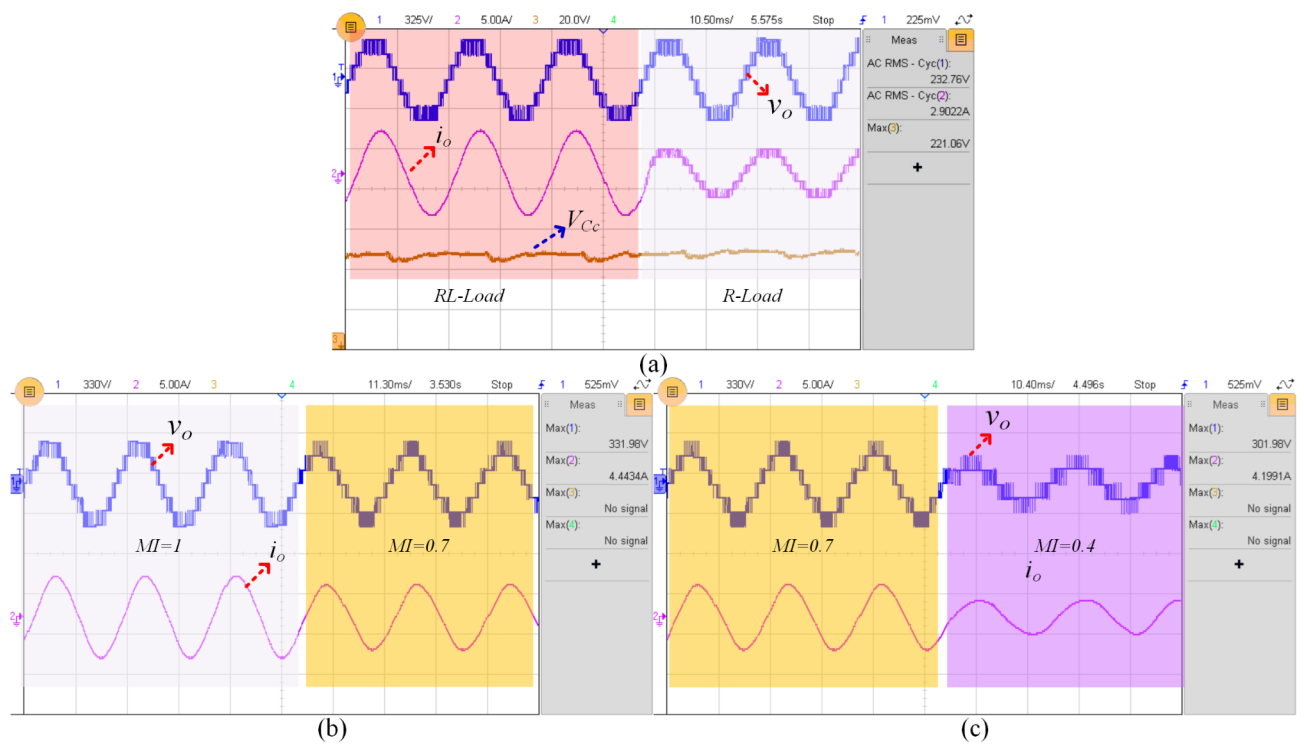


Fig. 13. Experimental results of proposed S2-7 L topology (a) v_o , i_o and V_{Cc} at load change from RL to R, and (c)&(d) v_o and i_o when MI = 1 to 0.7 and MI = 0.7 to 0.4.

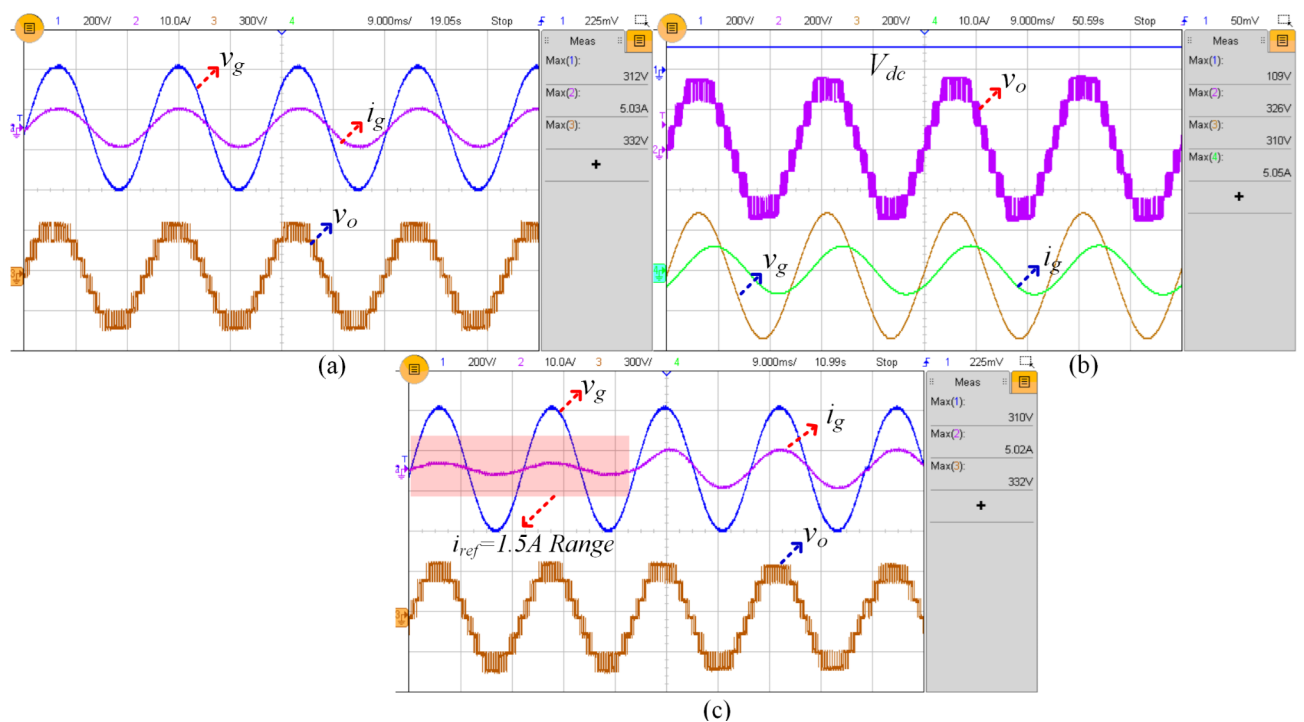


Fig. 14. Experimental results. (a) v_g (200 V/div), i_g (10 A/div) and v_o (300 V/div), (b) v_{dc} (200 V/div), v_o (200 V/div), v_g (200 V/div), and i_g (10 A/div), (c) v_g (200 V/div), i_g (10 A/div), and v_o (300 V/div) when i_{ref} is changed from 1.5 A to 3 A at UPF.

The FFT analysis of the proposed S2-7 L inverter topology was performed using MATLAB/Simulink software, and the corresponding results are presented in Fig. 15. As shown in Fig. 15 (a), the total harmonic distortion (THD) of the inverter output voltage is 15.33% at a modulation index of $m = 1$ and a switching frequency of 5 kHz. Additionally, the grid current THD is observed to be 2.85% as illustrated in Fig. 15 (b). When the modulation index is reduced to $m = 0.8$, the inverter output voltage decreases to 263.23 V with 7 L voltage steps. Additionally, the FFT profile was analyzed for this modulation index, revealing an inverter voltage THD of 21.45%, as shown in Fig. 15 (c).

Similarly, the FFT analysis of the proposed topology was verified experimentally, and the corresponding results are presented in Fig. 16. It can be observed that the experimental values of the inverter voltage THD and grid current THD are slightly higher i.e., 17.2% and 3.9%, than the simulation values but remain within the limits specified by IEEE-519 standards.

Power loss distribution for semiconductor switches is crucial for optimising the efficiency of multilevel inverters. Using PLECS (Plexim), power losses are analysed for various output power levels (100 W to 1000 W), focusing on individual semiconductor components such as IGBTs, and diodes. Figure 17 shows the radar chart that visually represents the power loss distribution across switches (P_1 – P_6) at different output powers (100–1000 W) and efficiency graph. It can be seen from the radar chart that, the switches P_4 , P_5 , P_1 , P_2 , and P_3 experience the most significant losses since they come in a capacitor charging path during $\pm V_{dc}$, $+V_{dc}$ and zero output voltage levels. The proposed S2-7 L inverter topology achieves a maximum measured efficiency of $\sim 96.7\%$ at 200 W, with total losses of ~ 7 W, closely matching the simulated efficiency of $\sim 97.9\%$. Figure 18 (h) presents the simulation and experimental efficiency trends across a power range of 100 W to 1000 W at 100 W intervals, showing that the experimental efficiency is 0.8–1.2% lower than the PLECS simulated efficiency.

Comparison study

This section provides a thorough comparison of the proposed S2-7 L inverter topology with other recent seven-level inverters that possess boosting abilities. A number of important aspects, including total power components (switch (S), gate driver (Dr), diode (D), capacitor (C) and inductor (I)), voltage boost (V_{BT}), voltage stress

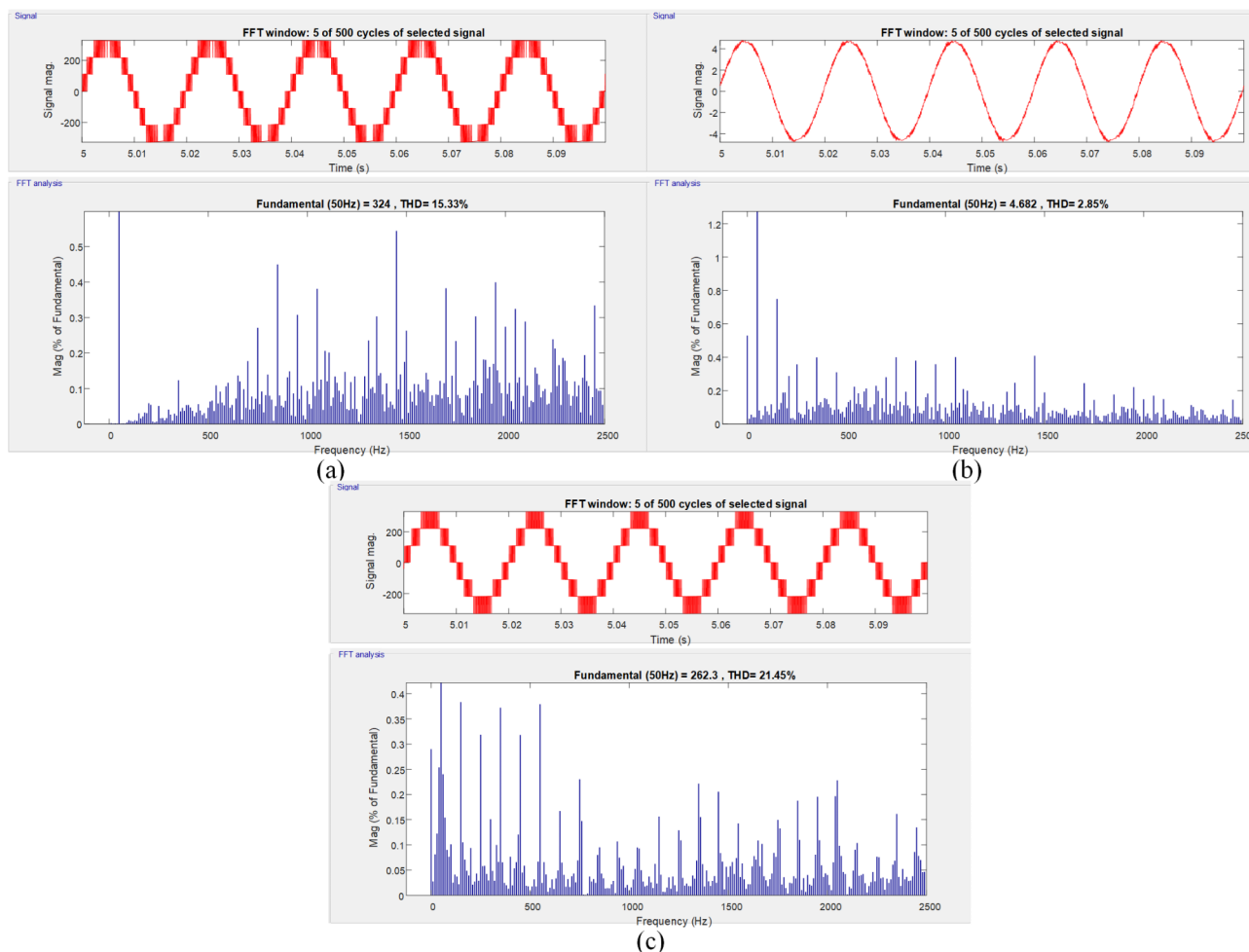


Fig. 15. Simulation results of (a) FFT waveform of inverter output voltage (V_o) at $m = 1$, (b) FFT waveform of grid current (i_g), (c) FFT waveform of inverter output voltage (V_o) at $m = 0.8$.

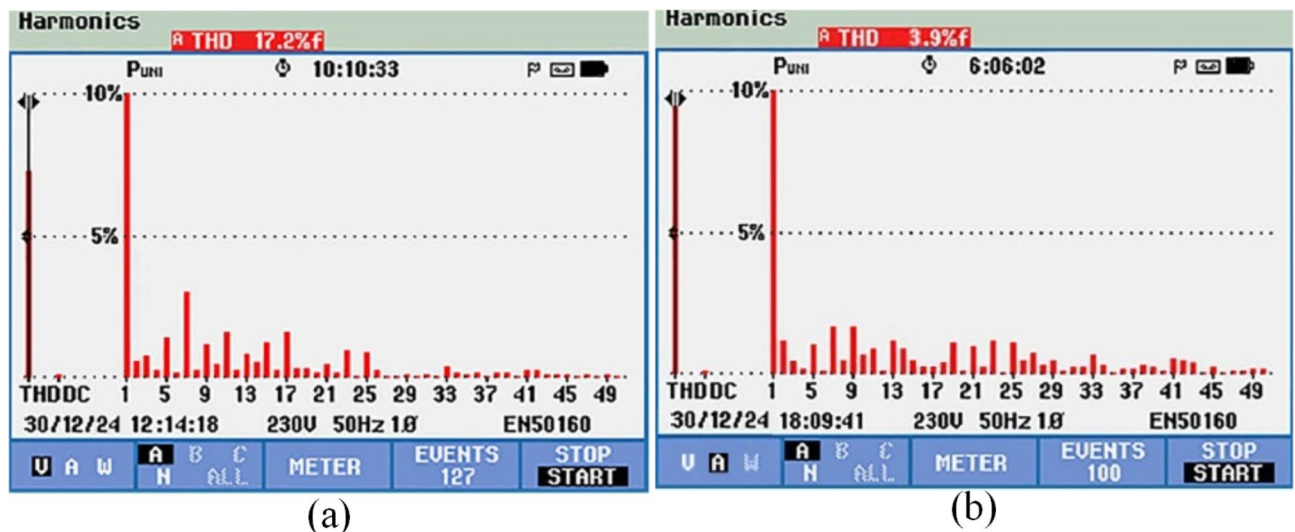


Fig. 16. Experimental THD results of (a) of inverter output voltage (V_o), (b) grid current (i_g)

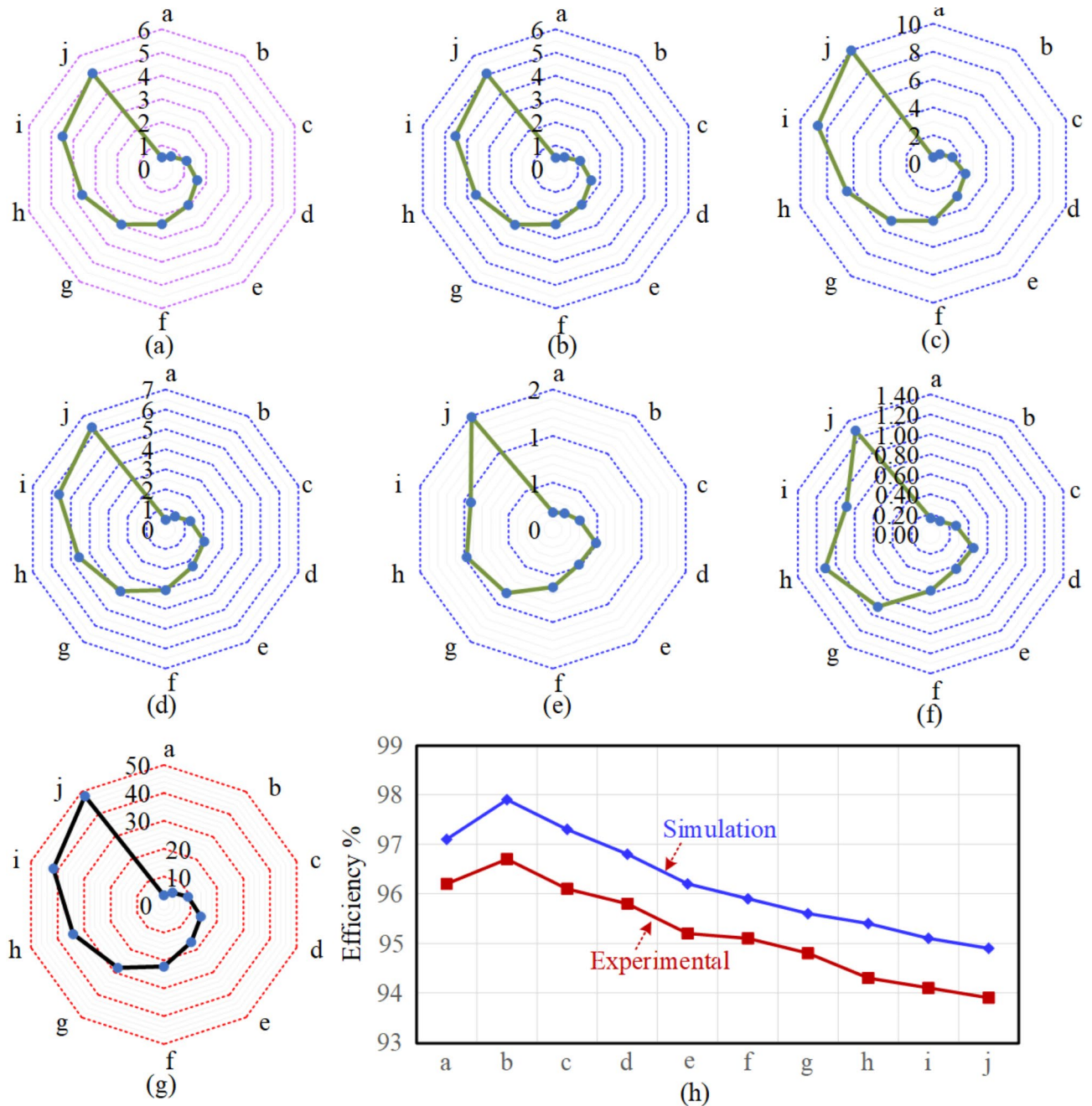
on switches ($V_{\text{stress-S}}$), total standing voltage per unit ($\text{TSV}_{\text{p.u.}}$), switches per Level (TS/L), maximum on state switches (M_{ons}), average of total conducting switches (A_M), diversity factor of capacitor (V_{DFC}), cost function (CF), number of switched with peak voltage stress (N_{PVS}), peak voltage stress of capacitor (V_{PSC}), H-bridge (HB), and efficiency (η), are evaluated throughout in order to point out the trade-offs between these parameters across different topologies.

In terms of power switches, and voltage boost (VBT)

Different seven-level inverter topologies have been proposed to achieve increased voltage levels while reducing component count, as detailed in Table 3. In the comparison of the proposed topology with other topologies that feature a reduced switch count, such as³⁰ with 7 switches and³¹ with 8 switches, it is clear that these configurations attain merely a boosting factor of 1.5. This limitation significantly restricts their ability to meet grid voltage requirements under low input voltage conditions. Even though the topologies like^{12,17,18} achieve the same boosting factor of 3 as the proposed topology, they require a higher switch count of 12. This increased switch count not only complicates the circuit design but also results in higher conduction losses, increased cost, and a larger overall system footprint. Other topologies proposed with boosting factors of 3, such as^{10,15,32}, require 10 switches, which is still significantly higher than the proposed topology's 6 switches. A reduced switch count directly translates to lower overall system cost, as power semiconductor devices are among the most expensive components in multilevel inverters. Furthermore, it minimizes conduction and switching losses, thereby enhancing the system's overall efficiency. Thus, while comparing all of the topologies listed in Table 3, the proposed S2-7 L topology is found to be the one that uses the fewest power switches i.e., the proposed S2-7 L topology uniquely combines a low switch count of 6 and a high boosting factor 3, offering a superior trade-off between efficiency, cost, reliability, and performance compared to other 7-level topologies. When comparing the total component count among topologies with a boosting factor of 3, it is observed that topologies such as^{11,15} utilize 21 components, while^{9,22,32} require 23 components, and^{6,8,10,20} use 24 components, and³³ use 25 components. Although the proposed S2-7 L topology employs a slightly higher number of switched capacitors to achieve a seven-level output voltage with triple voltage boost, it still maintains the lowest total power component count compared to all other topologies with the same boosting factor, demonstrating its superiority.

In terms of $\text{TSV}_{\text{p.u.}}$, TS/L , and M_{ons}

The $\text{TSV}_{\text{p.u.}}$ denotes the voltage stress experienced by each switch, normalized to the total output voltage. This metric is essential for evaluating the reliability and cost effectiveness of the inverter. The $\text{TSV}_{\text{p.u.}}$ of the proposed S2-7 L and the one proposed in³¹ is the least among all the topologies listed in Table 3, i.e., 4.67. The topology outlined in³⁰ has the second-lowest $\text{TSV}_{\text{p.u.}}$, followed by^{12,14,18,19,22}, which share the third position in the comparison Table 3, while all remaining topologies exhibit significantly higher $\text{TSV}_{\text{p.u.}}$. Although topologies^{14,19,30} exhibit a TSV comparable to the proposed seven-level inverter topology, they produce a seven-level output voltage with a boosting factor of only 1.5. In comparison, topologies^{12,18} also have comparable $\text{TSV}_{\text{p.u.}}$ with a three-fold voltage boost. However, the increased boosting capability in these topologies comes at the expense of higher switch counts and greater circuit complexity. Furthermore, the number of switches required per level (TS/L) in the proposed topology is the lowest among all the compared topologies. While the TS/L of the topologies presented in^{14,22,23,30,31,33} is comparable to that of the proposed topology, these topologies have a higher total power component count and $\text{TSV}_{\text{p.u.}}$ than the proposed S2-7 L topology. The proposed topology boasts the lowest TS/L ratio, a defining characteristic that distinguishes it from all other topologies. The maximum on-state switching devices in the proposed topology, as well as in^{30,31}, is 3, which is the lowest among the topologies listed in Table 3, resulting in reduced losses. However, the topologies in^{30,31}



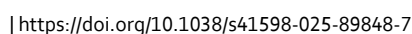
a:100W b: 200W c: 300W d:400W e:500W f:600W g:700W h:800W i:900W J:1000W

Fig. 17. PLEXIM loss details of (a)–(f) Power Switches P_1 – P_6 , (g) Total power loss including diode loss, and (h) Simulation and experimental efficiency Vs different output power.

lack the Common Ground (CG) feature, which is essential for effectively suppressing leakage current. Although the ANPC topology proposed in³¹ achieves some reduction in leakage current due to its circuit structure, the proposed S2-7 L topology offers maximum suppression of leakage current i.e., ~ 0 , making it more effective. The proposed S2-7 L topology stands out with the lowest TSVp.u., TS/L, and Mons among all compared topologies, showcasing its superiority.

Cost function (CF)

The cost function calculation is done using¹⁶ by the following expression,



nature portfolio 15

The CF of all the topologies has been computed using the Eq. (17) by considering different values of α , such as 0.5, 1, and 1.5 and are listed in Table 3. The cost function value of the proposed topology and the topologies proposed in^{11,15} are the greatest choices for lowering expenses since they both have the least and second least values compared to the other topologies in Table 3. With the best value of $T_{S/L}$ of 0.9, lowest M_{ONS} of 3, and least $TSV_{p,u}$ of 4.67, the proposed S2-7 L topology stands out as the better option. The combination of fewer M_{ONS} a lower $TSV_{p,u}$, and a reduced power components per Level $T_{S/L}$ in the proposed topology collectively reduces the losses and boosts overall efficiency significantly. Further, the proposed topology requires only 2 out of its 6 switches to be withstand the voltage more than output voltage and the remaining 4 switches are rated at V_{dr} and

[Ref]	Components					T _{NC}	Type	V _{BT}	V _{stress-s}							TSV _{p.u.} *	T _{S/L}	M _{ons}	A _M	V _{DFC}	CF with α			N _{pvs} (x V _{dc})	V _{pSC} (x V _{dc})	HB	η %	
	S	Dr	D	C	I				K	L	M	N	O	P	Q						0.5	1	1.5					
[7]	9	9	0	2	0	20	SC	1.5	5	2	2	0	0	0	0	7.5	1.3	5	4.14	0.67	15.8	18.3	20.8	2	0.5	N	97.18	
[13]	10	9	2	3	0	24	ANPC		1	9	0	0	0	0	0	6.2	1.4	5	4.29	1.33	18.1	20.1	22.2	7	1		92.12*	
[14]	8	8	2	4	0	22	SC		2	1	0	3	0	0	0	5.3	1.14	4	3	1.33	16.4	18.2	20.0	2	1		96.4*	
[16]	10	8	0	2	0	20			3	8	0	0	0	0	0	6.33	1.4	5	3.71	0.67	15.4	17.6	19.7	8	0.5		96.8	
[19]	10	8	0	3	0	21			4	6	0	0	0	0	0	5.3	1.4	6	3.71	1.33	15.8	17.5	19.3	6	2		NR	
[23]	8	8	2	2	0	20			0	3	5	0	0	0	0	7	1.14	4	3.42	0.67	15.7	18.0	20.3	5	0.5		98	
[30]	7	7	2	2	0	18			3	2	2	0	0	0	0	5	1	3	3	0.67	13.7	15.3	17.0	2	0.5		96.12	
[32]	8	8	2	3	0	21	ANPC		3	5	0	0	0	0	0	4.67	1.14	3	2	1.3	15.6	17.1	18.7	1.5	1		89*	
[5]	9	9	3	3	0	24	CGSC	3	0	6	0	0	3	0	0	6.67	1.3	5	4.14	2	9.1	10.2	11.3	3	3	Y	95.84*	
[6]	10	10	1	2	0	23			SC	0	4	0	6	0	0	0	6	1.4	5	4.71	1	8.7	9.7	10.7	6		2	97.63
[8]	10	10	0	2	1	23				0	4	0	2	4	0	0	6.67	1.4	4	3.14	0.67	8.8	9.9	11.0	4		2	97.3
[9]	10	9	1	1	1	22	0			2	0	2	5	0	0	7	1.4	5	4.57	0.67	8.5	9.7	10.8	5	2	NR		
[10]	10	8	2	3	0	23	CGSC		0	4	0	1	3	1	1	9	1.4	6	3.57	1.67	9.2	10.7	12.2	1	3	N	96.2	
[11]	9	9	1	2	0	21	SC		0	5	0	0	4	0	0	5.67	1.3	6	4	0.67	7.9	8.9	9.8	4	1	Y	93*	
[12]	12	11	0	2	1	26			0	8	0	4	0	0	0	5.33	1.7	7	6	0.67	9.6	10.4	11.3	4	1	N	98.7	
[15]	10	8	1	2	0	21			0	4	0	2	4	0	0	6.67	1.4	6	3.48	0.67	8.1	9.2	10.3	4	1	Y	95*	
[17]	12	10	4	4	0	30	TT		0	4	0	8	0	0	0	6.67	1.7	6	4.42	3.33	11.1	12.2	13.3	8	2		94.9*	
[18]	12	11	0	2	0	25	SC		0	8	0	4	0	0	0	5.3	1.7	7	6	0.67	9.2	10.1	11.0	4	2		N	93*
[20]	9	9	2	3	0	23			0	7	0	0	2	0	0	5.7	1.3	6	5.29	1	8.6	9.6	10.5	2	1	93.1		
[21]	10	10	4	2	0	26			0	4	0	4	2	0	0	6	1.4	4	3.57	0.67	9.7	10.7	11.7	2	1	94		
[22]	8	8	3	3	0	22	CGSC		0	3	0	0	3	0	0	5.3	1.14	4	3.14	1.5	8.2	9.1	10.0	3	2	Y	NR	
[31]	10	10	0	2	0	22			SC	0	4	0	2	0	4	0	6.67	1.43	6	4.86	0.67	8.4	9.6	10.7	4		1	98.3
[33]	12	11	0	2	0	25				0	8	0	4	0	0	0	5.33	1.7	7	7	0.33	9.2	10.1	11	4		0.5	N
[Pro]	6	6	4	4	0	20	CGSC		0	2	0	2	0	2	0	4.67	0.9	3	2.85	2	7.4	8.2	9.0	2	2	N	97.9	

Table 3. Comparative analysis. K: $V_{dc}/2$, L: V_{dc} , M: $3V_{dc}/2$, N: $2V_{dc}$, O: $3V_{dc}$, P: $4V_{dc}$, Q: $>5V_{dc}$, Y: Yes, N: No, A_m : Average of total conducting switches, N_{pvs} : Number of switches with peak voltage stress, V_{pSC} : Peak voltage stress of the switched capacitor, V_{bt} : Boost factor, *: Measured efficiency, NR: Not reported, CGSC: Common Ground Switched Capacitor, SC: Switched Capacitor, TT: T-Type

$2V_{dc}$. This lowers the overall cost of the system in comparison with all other topologies. Figure 18 shows the merits of the proposed topology against other 7 L inverter topologies.

Conclusion

A seven-level SC-based triple boost transformerless inverter with a common-ground structure, utilizing a minimal number of switches, has been proposed in this paper. The proposed S2-7 L common ground topology eliminates leakage current by directly connecting the grid's neutral to the PV panels' negative terminal, ensuring suitability for transformerless grid-tied solar PV systems. Even though the maximum voltage stress on the switch is significant, the TSVp.u. is lower than in other topologies, making the proposed topology more cost-effective and efficient due to fewer components and reduced power loss. Unlike other topologies, the proposed design achieves a low TSVp.u. even with fewer devices, breaking the usual trade-off between switch count and TSVp.u. The experimental tests using 825 W rated power laboratory prototype have been carried out, and the results are presented to verify the effectiveness of the proposed S2-7 L topology. The proposed S2-7 L inverter topology is proven superior through a detailed comparison with other 7 L topologies. Furthermore, it has an increased efficiency of 97.9% and a low cost, indicating that the proposed topology is a potential candidate for grid-tied PV applications.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

Received: 15 November 2024; Accepted: 7 February 2025

Published online: 14 March 2025

References

1. Salem, A., Van Khang, H., Robbersmyr, K. G., Norambuena, M. & Rodriguez, J. Voltage source multilevel inverters with reduced device count: Topological review and Novel comparative factors. *IEEE Trans. Power Electron.* **36**, 2720–2747 (2021).
2. Hinago, Y. & Koizumi, H. A switched-capacitor inverter using series/parallel conversion with inductive load. *IEEE Trans. Ind. Electron.* **59**, 878–887 (2012).

3. Pandurangan, G. N., Krishnasamy, V., Ali, J. S. M., Mahmood, F. M. & Almakhlles, D. Single-phase common ground type 5L inverter with reduced capacitor voltage stress for photovoltaic applications. *IET Power Electron.* **16**, 883–892 (2023).
4. Anand, V., Singh, V. & Mohamed Ali, J. S. Dual boost five-level switched-capacitor inverter with common ground. *IEEE Trans. Circuits Syst. II Express Briefs.* **70**, 556–560 (2023).
5. Jakhar, A., Sandeep, N. & Verma, A. K. Seven-level common-ground-type inverter with reduced voltage stress. *IEEE J. Emerg. Sel. Top. Power Electron.* **12**, 2108–2115 (2024).
6. Jakhar, A., Sandeep, N. & Verma, A. K. Common-ground-type inverter topology with low voltage stress and boosting ability. *IEEE Trans. Circuits Syst. II Express Briefs.* **71**, 2854–2858 (2024).
7. Prasadarao, V. S. K., Peddapaty, S. & Kumar, B. A. Voltage-boosting seven-level switched capacitor multilevel inverter with reduced device Count. *IEEE J. Emerg. Sel. Top. Power Electron.* **12**, 743–753 (2024).
8. Kumari, S. & N, S. Switched-capacitor-based seven-level inverter with reduced component count and current stress. *IEEE J. Emerg. Sel. Top. Ind. Electron.* **5**, 2–7 (2023).
9. Shahsavari, T. H., Babaei, E., Sabahi, M. & khoshkbar-Sadigh, A. A new boost type single source seven-level switched-capacitor based inverter with reduced current stress over the components. *Comput. Electr. Eng.* **101**, 108160 (2022).
10. Srivastava, A. & Seshadrinath, J. A. Single-phase seven-level triple boost inverter for grid-connected transformerless PV applications. *IEEE Trans. Ind. Electron.* **70**, 9004–9015 (2023).
11. Chen, J., Wang, C. & Li, J. A. Single-phase step-up seven-level inverter with a simple implementation method for level-shifted modulation schemes. *IEEE Access.* **7**, 146552–146565 (2019).
12. Singh, A. K., Mandal, R. K. & Anand, R. Quasi-resonant switched-capacitor-based seven-level inverter with reduced capacitor spike current. *IEEE J. Emerg. Sel. Top. Power Electron.* **11**, 1953–1965 (2023).
13. Cardoso, R. D. B., Da Silva, E. R. C., Limongi, L. R. & Costa, A. E. L. D. C. A seven-level inverter with natural balance and boosting capability. *IEEE Trans. Ind. Appl.* **59**, 925–937 (2023).
14. Kurdkandi, N. V. et al. A new seven-level transformer-less grid-tied inverter with leakage current limitation and voltage boosting feature. *IEEE J. Emerg. Sel. Top. Ind. Electron.* **4**, 228–241 (2022).
15. Kishore, P. S. V. et al. A new reduced switch seven-level triple boost switched capacitor based inverter. *IEEE Access.* **10**, 73931–73944 (2022).
16. Zhao, J., Chen, Y., Zeng, J. & Liu, J. Low-voltage stress seven-level inverter based on symmetrical capacitors. *IEEE J. Emerg. Sel. Top. Power Electron.* **10**, 3033–3044 (2022).
17. Chen, M. Novel cascaded seven-level inverter with embedded voltage boosting for renewable energy applications. *CPSS Trans. Power Electron. Appl.* **7**, 58–70 (2022).
18. Siddique, M. D. et al. A new switched capacitor 7L inverter with triple voltage gain and low voltage stress. *IEEE Trans. Circuits Syst. II Express Briefs.* **67**, 1294–1298 (2020).
19. Liu, J., Zhu, X. & Zeng, J. A. Seven-level Inverter with Self-Balancing and low-voltage stress. **8**, 685–696 (2020).
20. Roy, T., Tesfay, M. W., Nayak, B. & Panigrahi, C. K. A 7-Level switched capacitor multilevel inverter with reduced switches and voltage stresses. *IEEE Trans. Circuits Syst. II Express Briefs.* **68**, 3587–3591 (2021).
21. Siddique, M. D. et al. Reduced switch count based single source 7L boost inverter topology. *IEEE Trans. Circuits Syst. II Express Briefs.* **67**, 3252–3256 (2020).
22. Grigoletto, F. B., de Vilhena Moura, P. H. M., Chaves, D. B. & Vilaverde, J. D. S. Step-up seven-level common-ground transformerless inverter. In *14th IEEE Int. Conf. Ind. Appl. INDUSCON 2021 - Proceedings* 716–722 <https://doi.org/10.1109/INDUSCON51756.2021.9529417> (2021).
23. Reddy, B. P. et al. 7L-SCBI topology with minimal semiconductor device count. *IET Power Electron.* **13**, 3065–3071 (2020).
24. Gopinath, N. P., Vijayakumar, K., Ali, M., Raghupathi, J. S. & Selvam, S. K. A Triple boost seven-level common ground transformerless inverter topology for grid-connected photovoltaic applications. *Energies* **16**, (2023).
25. Almakhlles, D. J. & Sathik, M. J. Single – phase transformerless nine – level inverter with voltage boosting ability for PV fed AC microgrid applications. *Sci. Rep.* 1–11. <https://doi.org/10.1038/s41598-022-16057-x> (2022).
26. Narayanan Pandurangan, G. & Vijayakumar, K. Dual-ground transformerless nine-level dynamic voltage boosting inverter topology. *Int. J. Electron.* **111**, 677–696 (2024).
27. Jakhar, A., Sandeep, N. & Verma, A. K. Common-ground-type quadruple boosting nine-level inverter. *IEEE J. Emerg. Sel. Top. Ind. Electron.* **PP**, 1–8 (2024).
28. Anand, V. & Singh, V. A 13-level switched-capacitor multilevel inverter with single DC source. *IEEE J. Emerg. Sel. Top. Power Electron.* **10**, 1575–1586 (2022).
29. Islam, S., Siddique, M. D., Iqbal, A., Mekhilef, S. & Al-Hitmi, M. A switched capacitor-based 13-level inverter with reduced switch count. *IEEE Trans. Ind. Appl.* **58**, 7373–7383 (2022).
30. Lin, X., Yan, Z. J. F. B., Liu, J. & Z. Switched-capacitor based seven-level Boost Inverter with a reduced number of devices. *CSEE J. Power Energy Syst.* **10**, 381–391 (2024).
31. Che, A. S., Abro, Y., Zafar, A. & G.E. M & Design analysis of seven-level active neutral point clamped inverter based on diode and switched technique. *Arab. J. Sci. Eng.* <https://doi.org/10.1007/s13369-024-09725-9> (2024).
32. Mondal, S., Biswas, S., Islam, M. R., Hossain, M. K. & Raad, R. A. Seven-level switched-capacitor based transformerless inverter with modified PWM strategy to enhance the performance of grid-connected PV systems. *IET Power Electron.* **17**, 855–868 (2024).
33. Meraf, S. T. et al. A diamond shaped multilevel inverter with dual mode of operation. *IEEE Access.* **9**, 59873–59887 (2021).

Author contributions

NPG, RA, MJS, DK: Conceptualization, Methodology, Software, Visualization, Investigation, Writing- Original draft preparation. NPG, MJS: Data curation, Validation, Supervision, Resources, Writing—Review & Editing. NPG, RA, MJS: Project administration, Supervision, Resources, Writing—Review & Editing.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to M.J.S.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2025